

Manoj Kumar Kola

Senior Data Engineer

CONTACT

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EDUCATION

**Master of Science:
Data Science**
Bellevue University, NE
Expected 08/2026

**B.Tech: Electrical &
Electronics Engineering**
JNTUH, India
05/2008

CERTIFICATIONS

Leadership Essentials
McKinsey & Company
Business · Team · Self Leadership

Data Science with SQL & Tableau
Cornell University

Data Visualization in Tableau
Cornell University

PROFESSIONAL SUMMARY

Senior Data Engineer with 16+ years of progressive experience designing, building, and operating enterprise-grade ETL/ELT pipelines across financial services and healthcare domains. Proven technical lead with 8+ years guiding cross-functional teams — most recently serving as Technical Anchor across a 30-person, 5-team engineering organization at Discover Financial Services. Hands-on expertise in Databricks, PySpark, SQL, Python, and Snowflake, with deep experience delivering scalable lakehouse architectures, medallion-pattern pipelines, and curated data products that power analytics and operational decision-making. Healthcare data experience includes ETL pipeline development for UnitedHealth Group's Medicare program. Currently pursuing MS in Data Science (Bellevue University, expected 08/2026).

PROFESSIONAL EXPERIENCE

Capital One Principal Application Engineer	05/2025 – Present Remote, OH
- Architected and maintain 4–5 enterprise-grade ELT pipelines on Databricks supporting the Discover–Capital One Backbook migration, integrating customer, account, and transaction data across alerts and spend controls — with continuous enhancements ensuring scalability and reliability at enterprise scale. - Designed medallion-pattern (bronze/silver/gold) data pipelines ingesting from databases, APIs, and flat files — transforming and standardizing records to align with unified data models across merged entities. - Implemented data validation, reconciliation, and quality checks across integration pipelines, ensuring accuracy and compliance with governance standards through SIT/UAT and production deployments. - Defined data contracts and integration requirements with cross-functional engineering and business teams, translating business needs into reliable, production-grade data products. - Monitored and optimized Databricks pipeline performance for reliability and scalability as data volumes and downstream consumer demands grew post-merger.	
Discover Financial Services Application Engineer — Technical Anchor	01/2021 – 04/2025 Remote, OH
- Served as Technical Anchor across a 30-person, 5-team Recovery Systems organization — directly leading a team of 6 engineers while providing architecture guidance, code review, and cross-team technical support. - Designed and delivered real-time and batch data ingestion frameworks on Databricks using PySpark, standardizing data flow and enabling self-service access to curated, trusted datasets for analytics consumers. - Translated complex business and technology requirements into scalable ETL/ELT architecture using Agile methodology, accelerating delivery of high-priority recovery data solutions. - Established data governance and quality standards — enforcing validation, reconciliation, and SLA compliance checks that reduced data integrity issues reaching downstream consumers. - Identified technical improvement opportunities across existing pipelines, driving gains in maintainability, observability, and engineering team productivity.	
TATA Consultancy Services — Nationwide Insurance Technical Lead	02/2020 – 01/2021 Columbus, OH
- Led development of business rule transformation logic for Nationwide's commercial lines using IBM ODM and Eclipse Rule Designer, converting complex business requirements into executable technical specifications. - Managed offshore engineering team via daily Agile scrum — allocating tasks by priority, reviewing rule changes, and merging Git branches to ensure on-time delivery and code quality. - Deployed pipeline and rule changes through structured Git-based CI/CD workflows, reducing release risk and improving deployment consistency. - Validated business rule logic end-to-end using SOAP UI 5.0 before each production promotion, ensuring functional accuracy and stakeholder sign-off compliance.	

SKILLS

CORE ENGINEERING

Python · PySpark · SQL · PL/SQL

PLATFORMS & CLOUD

Databricks · Snowflake · AWS · Azure (in progress)

ETL & PIPELINE TOOLS

Informatica PowerCenter · Ab Initio · dbt (in progress)

DATA ARCHITECTURE

Medallion · Kimball · Star Schema · SCD Type 1 · Lakehouse

DATA QUALITY

Validation · Reconciliation · SLA Monitoring · Lineage · Observability

DATABASES

IBM DB2 · MS SQL Server · Oracle

DEVOPS & VCS

Git · BitBucket · SVN · CI/CD Pipelines

REPORTING

Tableau · IBM Cognos 10

SCHEDULERS

Autosys · Control-M · Maestro

STRENGTHS

- Willingness to learn new things, positive attitude and quick learner.
- Committed and adaptive to new technologies and environments.
- Good communication skills, balanced approach with team spirit.
- Versatile and highly motivated with discipline and punctuality.

PERSONAL INFORMATION

Gender: Male

Ethnicity: Asian

Nationality: Indian

Visa Status: H1B

Marital Status: Married

TATA Consultancy Services — JP Morgan Chase

05/2016 – 12/2019

ETL Lead Developer

Newark, DE

- Led ETL development for JP Morgan Chase data warehouse using Informatica PowerCenter, building mappings and workflows aligned to Kimball dimensional modeling and SCD Type 1 patterns.
- Extracted and staged data from multiple operational source systems into star schema data warehouse structures, supporting analytics, reporting, and downstream business consumers.
- Prepared DDL change scripts, stored procedure updates, and promotion documents; guided team through complex pipeline troubleshooting and ensured delivery within SLA commitments.
- Reviewed team mappings, sessions, and workflow changes to ensure alignment with client requirements and data standards prior to each release.

UnitedHealth Group

07/2014 – 05/2016

Sr. Informatica Developer — Medicare Domain

- Engineered ETL pipelines for UnitedHealth Group's Medicare/PBC healthcare domain using Informatica PowerCenter and IBM DB2, processing member benefits, claims, and provider data to support clinical reporting and analytics.
- Translated healthcare business requirements into technical ETL specifications, ensuring accurate data population across Medicare subject areas in a regulated environment.
- Developed and maintained Informatica mappings and workflows across multiple subject areas; extracted source data and loaded staging area, data warehouse, and data marts using SCD Type 1 methodology.
- Generated operational reports using IBM Cognos 10, delivering insights on Medicare program performance to business users and onsite specialists.
- Guided team members through complex work requests, allocating tasks by criticality and reviewing code and documentation to uphold client data quality standards.

Capgemini — Liberty Mutual Insurance

10/2013 – 06/2014

Team Lead

- Led an offshore ETL development team delivering Informatica PowerCenter solutions for Liberty Mutual, managing task allocation by priority while ensuring code quality and on-time delivery.
- Translated business requirements from onsite stakeholders into technical specifications, bridging communication between business users and the offshore engineering team.
- Developed and modified Informatica mappings and workflows; prepared DDL change scripts, design documents, and promotion documentation for each release.
- Reviewed and approved code and documents prepared by team members, ensuring alignment with client data standards before production deployment.

Tech Mahindra — Selective Insurance

02/2010 – 10/2013

Informatica Developer

- Designed and developed Informatica PowerCenter mappings, sessions, and workflows to extract, transform, and load policy and claims data into the data warehouse and data marts using SCD Type 1 methodology.
- Built ETL pipelines across multiple subject areas — policy, claims, and financial data — ensuring accuracy and integrity throughout the staging, warehouse, and data mart layers.
- Prepared design documents, DDLs, SQL validation scripts, and deployment documentation; tested ETL code end-to-end and tracked defects in HPSM through to resolution.
- Monitored production jobs and provided timely resolution during pipeline failures, maintaining SLA commitments and minimizing business disruption.